

Revised Generation Interconnection System Impact Study Report

For Queue Project Q11-0066

Dooville 230 kV
May 26, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0066. The Interconnection Customer proposes a 64.6 MW battery energy storage facility interconnecting at the Dooville 230 kV substation in the Indiana area. The request was received on August 25, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 54 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$27,319,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Swan Valley-Twin County Ranch 230 KV	Swan Valley-Trego 500 KV	129.8	439	10.3
Monfort-Wabash 230 KV	Montpelier Electric Generating Station Substation-Waynedale 230 KV	109.6	818	10.6
Waynedale-Montpelier Electric Generating Sta 230 KV	Wabash 500 kV/230 kV Transformer Bank	108.5	809	20.1
Dooville-Thomas Hill Sta 230 KV	Montpelier Electric Generating Sta-Wabash 500 KV	103.5	578	46.2
Dooville-Montpelier Electric Generating Sta 230 KV Line	Montpelier Electric Generating Station-Wabash 500 KV	102.9	562	53.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Swan Vly-Twin County Ranch 230 KV	Rebuild line with higher-capacity conductor	\$32,171,000	6% (\$2,045,000)
Monfort-Wabash 230 KV	Reconductor line	\$13,363,000	4% (\$465,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Waynedale-Montpelier Electric Generating Sta 230 KV	Reconductor line	\$30,326,000	6% (\$1,847,000)
Dooville-Thomas Hill Sta 230 KV	Reconductor line	\$11,176,000	100% (\$11,176,000)
Dooville-Montpelier Electric Generating Sta 230 KV Line	Reconductor line	\$11,786,000	100% (\$11,786,000)
	Total allocated to project		\$27,319,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.